

Treatment of Pattern Hair Loss

The latter group is important to identify, as these patients are unlikely to be satisfied with treatments aimed at restoring hair growth, and involvement of other healthcare professionals—such as a clinical psychologist—may be necessary. Treatments for pattern hair loss are available, but patients should be counseled that none will restore hair growth to its prepubertal density. While some recovery of hair growth is possible, the primary goal of medical treatments is to prevent further progression (see Box 86-2).

Men

Two medical treatments—**topical minoxidil solution (TMS)** and **oral finasteride**—are licensed for the treatment of male pattern hair loss and can be used in combination. A trial of at least 6 months is required to assess efficacy, and treatment must be continued indefinitely to maintain results.

Box 86-2: Treatment of Androgenetic Alopecia

General Principle:

Caution against unrealistic expectations; the primary goal is to halt disease progression.

Male Pattern Hair Loss: - **5% topical minoxidil solution or foam**, applied twice daily

- **Oral finasteride**, 1 mg daily

- **Combination therapy:** Use for at least 6 months to assess response; continue treatment to maintain benefits

- **Surgery (e.g., hair transplantation):** Most effective for restoring frontal hair loss

Female Pattern Hair Loss: - **2%–5% topical minoxidil solution**, applied twice daily

- **Topical 17 β -estradiol** (e.g., estradiol benzoate, 20–25 mg/mL in isopropanol)

- **Oral anti-androgens** (e.g., spironolactone, cyproterone acetate)
- **Combination therapy:** Use for at least 6 months to assess response; continue treatment to maintain benefits
- Maintain **serum ferritin** $>40 \mu\text{g/L}$
- **Hair transplantation** in selected cases

MINOXIDIL

Clinical trials using various endpoints—including hair counts, hair weight, and global photography—have shown that up to 60% of men experience improvement in vertex balding with 5% topical minoxidil, with a mean increase in hair density of 10% to 12%. Response rates with 2% minoxidil are lower. Treatment must be continued to sustain results; any gains are typically lost within 4 to 6 months after discontinuation.

Minoxidil is not effective for frontotemporal recession. Scalp irritation may occur, most commonly due to propylene glycol in the formulation.

FINASTERIDE

Finasteride is a type 2 5α -reductase inhibitor. At a dose of 1 mg daily, it prevents progression of hair loss in most men. After 2 years of treatment, approximately two-thirds show some improvement in hair growth. In long-term studies, a slight decline in hair count was observed after 5 years, but hair loss progressed more rapidly in untreated individuals.

A small but statistically significant increase in sexually related adverse effects—such as impotence—has been reported; however, finasteride is otherwise well-tolerated. Any benefit is lost within 12 months of stopping treatment.

The exact mechanism of action in responders remains unclear. Recent evidence suggests that finasteride may reactivate "hypotrophic" terminal hair follicles by promoting faster entry into and prolonging the anagen phase, rather than inducing true vellus-to-terminal hair conversion.

Despite a lack of formal clinical trials, **combination therapy with topical minoxidil and oral finasteride** is commonly used in practice. Dual inhibition

of both type 1 and type 2 5 α -reductase may offer additional benefit in treating male pattern hair loss.

SURGERY

Various surgical techniques exist for male pattern hair loss, with **hair transplantation** being the most widely used. These procedures redistribute existing terminal hairs to balding areas, without increasing the total number of terminal hairs. When performed by experienced surgeons and with appropriate patient selection, excellent cosmetic outcomes can be achieved.

Surgery is the only treatment capable of improving hair coverage on completely bald scalp and can rapidly restore hair. It is particularly effective for frontal hair loss (see Chap. 250).

Women

(Continued section—content likely follows on female pattern hair loss, including specific considerations for diagnosis, treatment response, and management strategies.)